RF IS DEFINED AT A DIRECT SUM EXACTLY WHEN DEFINED AT BOTH SUMMANDS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 5031-5109, following the statement at 5018-5029.

Editorial note. The closing 'The proof of the corresponding statements for LF are dual' (line 5108) is a mirror-level omission, recorded.

Allowed standard prerequisites (editorial).

- Split triangles. lemma-split, cited at proof lines 5033 and 5038; complete proof at source lines 916-927, queued for separate carding. Supplies the split triangle X -> X + Y -> Y -> X[1] and the identification of the middle triangle with a direct sum.
- Two-out-of-three for definedness. lemma-2-out-of-3-defined (carded in this batch, L26162), cited at proof line 5034. Supplies that RF defined at X and Y implies defined at X + Y with the split triangle.
- Arrows inverted by localization. categories-lemma-what-gets-inverted, cross-chapter citation at proof line 5047 (tags mechanism). Supplies that saturated S is exactly the set of arrows inverted by Q, giving s + s' in S.
- Filteredness of slice categories and products. categories-remark-left-localization-morphisms-colimit and categories-lemma-essentially-constant-over-product, cross-chapter citations at proof lines 5055-5058 (tags mechanism). Supplies X/S, Y/S, (X + Y)/S filtered and the product criterion for essential constancy.
- Direct sums of essentially constant systems. homology-lemma-direct-sum-from-product-essentially-constant, cross-chapter citation at proof line 5070 (tags mechanism; Karoubian hypothesis enters here). Supplies that essential constancy of the direct sum system implies essential constancy of both summand systems.

Source reference. Original reference 04E6: focused-seeds/geometry/S9-der-direct-sum-defined/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0A1S: focused-seeds/geometry/S9-der-direct-sum-defined/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 0A2C: focused-seeds/geometry/S9-der-direct-sum-defined/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 05Q9: focused-seeds/geometry/S9-der-direct-sum-defined/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 05Q0: focused-seeds/geometry/S9-der-direct-sum-defined/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0A2G: focused-seeds/geometry/S9-der-direct-sum-defined/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05SC: sources/stacks-derived/pinned-original-source.tex, source lines 4896–4896. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05QT: sources/stacks-derived/pinned-original-source.tex, source lines 917–917. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 05S8: sources/stacks-derived/pinned-original-source.tex, source lines 4736–4736. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Assumptions and notation as in Situation 05S8. Let X, Y be objects of $\mathcal{D}$.*

(1) *If RF is defined at X and Y, then RF is defined at $X \oplus Y$.*
(2) *If $\mathcal{D}'$ is Karoubian and RF is defined at $X \oplus Y$, then RF is defined at both X and Y.*

In either case we have $RF(X \oplus Y) = RF(X) \oplus RF(Y)$. Similarly for LF.

Proof. If RF is defined at X and Y, then the distinguished triangle $X \to X \oplus Y \to Y \to X[1]$ (Lemma 05QT) and Lemma 05SC shows that RF is defined at $X \oplus Y$ and that we have a distinguished triangle $RF(X) \to RF(X \oplus Y) \to RF(Y) \to$

$RF(X)[1]$. Applying Lemma 05QT to this once more we find that $RF(X \oplus Y) = RF(X) \oplus RF(Y)$. This proves (1) and the final assertion.

Conversely, assume that RF is defined at $X \oplus Y$ and that $\mathcal{D}'$ is Karoubian. Since S is a saturated system S is the set of arrows which become invertible under the additive localization functor $Q : \mathcal{D} \to S^{-1}\mathcal{D}$, see Categories, Lemma 05Q9. Thus for any $s : X \to X'$ and $s' : Y \to Y'$ in S the morphism $s \oplus s' : X \oplus Y \to X' \oplus Y'$ is an element of S. In this way we obtain a functor

$$X/S \times Y/S \longrightarrow (X \oplus Y)/S$$

Recall that the categories $X/S, Y/S, (X \oplus Y)/S$ are filtered (Categories, Remark 05Q0). By Categories, Lemma 0A2C $X/S \times Y/S$ is filtered and $F|_{X/S} : X/S \to \mathcal{D}'$ (resp. $G|_{Y/S} : Y/S \to \mathcal{D}'$) is essentially constant if and only if $F|_{X/S} \circ \mathrm{pr}_1 : X/S \times Y/S \to \mathcal{D}'$ (resp. $G|_{Y/S} \circ \mathrm{pr}_2 : X/S \times Y/S \to \mathcal{D}'$) is essentially constant. Below we will show that the displayed functor is cofinal, hence by Categories, Lemma 0A1S, we see that $F|_{(X \oplus Y)/S}$ is essentially constant implies that $F|_{X/S} \circ \mathrm{pr}_1 \oplus F|_{Y/S} \circ \mathrm{pr}_2 : X/S \times Y/S \to \mathcal{D}'$ is essentially constant. By Homology, Lemma 0A2G (and this is where we use that $\mathcal{D}'$ is Karoubian) we see that $F|_{X/S} \circ \mathrm{pr}_1 \oplus F|_{Y/S} \circ \mathrm{pr}_2$ being essentially constant implies $F|_{X/S} \circ \mathrm{pr}_1$ and $F|_{Y/S} \circ \mathrm{pr}_2$ are essentially constant proving that RF is defined at X and Y.

Proof that the displayed functor is cofinal. To do this pick any $t : X \oplus Y \to Z$ in S. Using MS2 we can find morphisms $Z \to X'$, $Z \to Y'$ and $s : X \to X'$, $s' : Y \to Y'$ in S such that

$$\begin{array}{ccccc} X & \longleftarrow & X \oplus Y & \longrightarrow & Y \\ \downarrow{\scriptstyle s} & & \downarrow & & \downarrow{\scriptstyle s'} \\ X' & \longleftarrow & Z & \longrightarrow & Y' \end{array}$$

commutes. This proves there is a map $Z \to X' \oplus Y'$ in $(X \oplus Y)/S$, i.e., we get part (1) of Categories, Definition 04E6. To prove part (2) it suffices to prove that given $t : X \oplus Y \to Z$ and morphisms $s_i \oplus s_i' : Z \to X_i' \oplus Y_i'$, $i = 1, 2$ in $(X \oplus Y)/S$ we can find morphisms $a : X_1' \to X'$, $b : X_2' \to X'$, $c : Y_1' \to Y'$, $d : Y_2' \to Y'$ in S such that $a \circ s_1 = b \circ s_2$ and $c \circ s_1' = d \circ s_2'$. To do this we first choose any X' and Y' and maps a, b, c, d in S; this is possible as X/S and Y/S are filtered. Then the two maps $a \circ s_1, b \circ s_2 : Z \to X'$ become equal in $S^{-1}\mathcal{D}$. Hence we can find a morphism $X' \to X''$ in S equalizing them. Similarly we find $Y' \to Y''$ in S equalizing $c \circ s_1'$ and $d \circ s_2'$. Replacing X' by X'' and Y' by Y'' we get $a \circ s_1 = b \circ s_2$ and $c \circ s_1' = d \circ s_2'$.

The proof of the corresponding statements for LF are dual. □